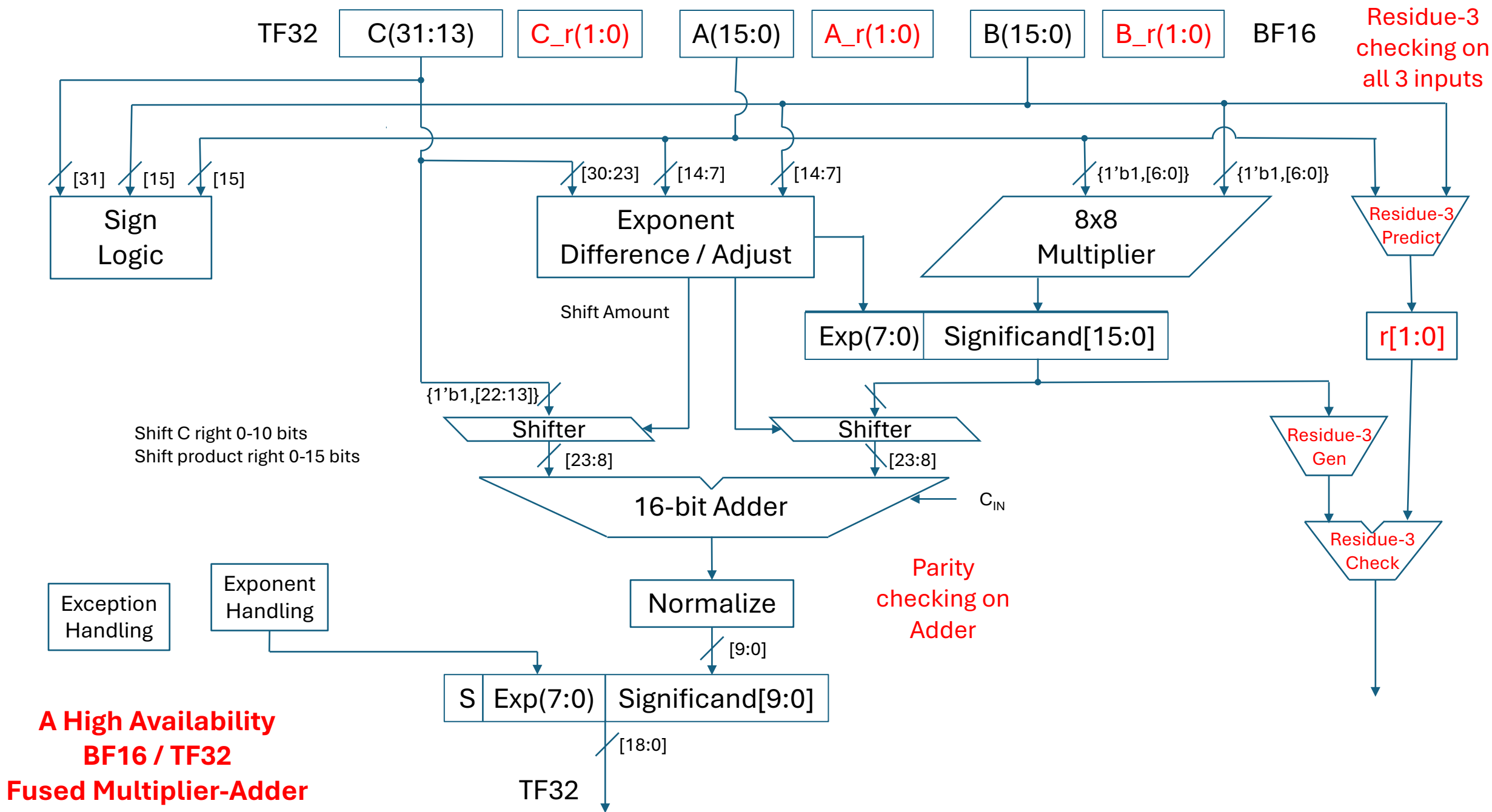


A High Availability BF16 / TF32 Fused Multiplier-Adder

Donald McCauley TechAnalytics LLC March 14, 2026



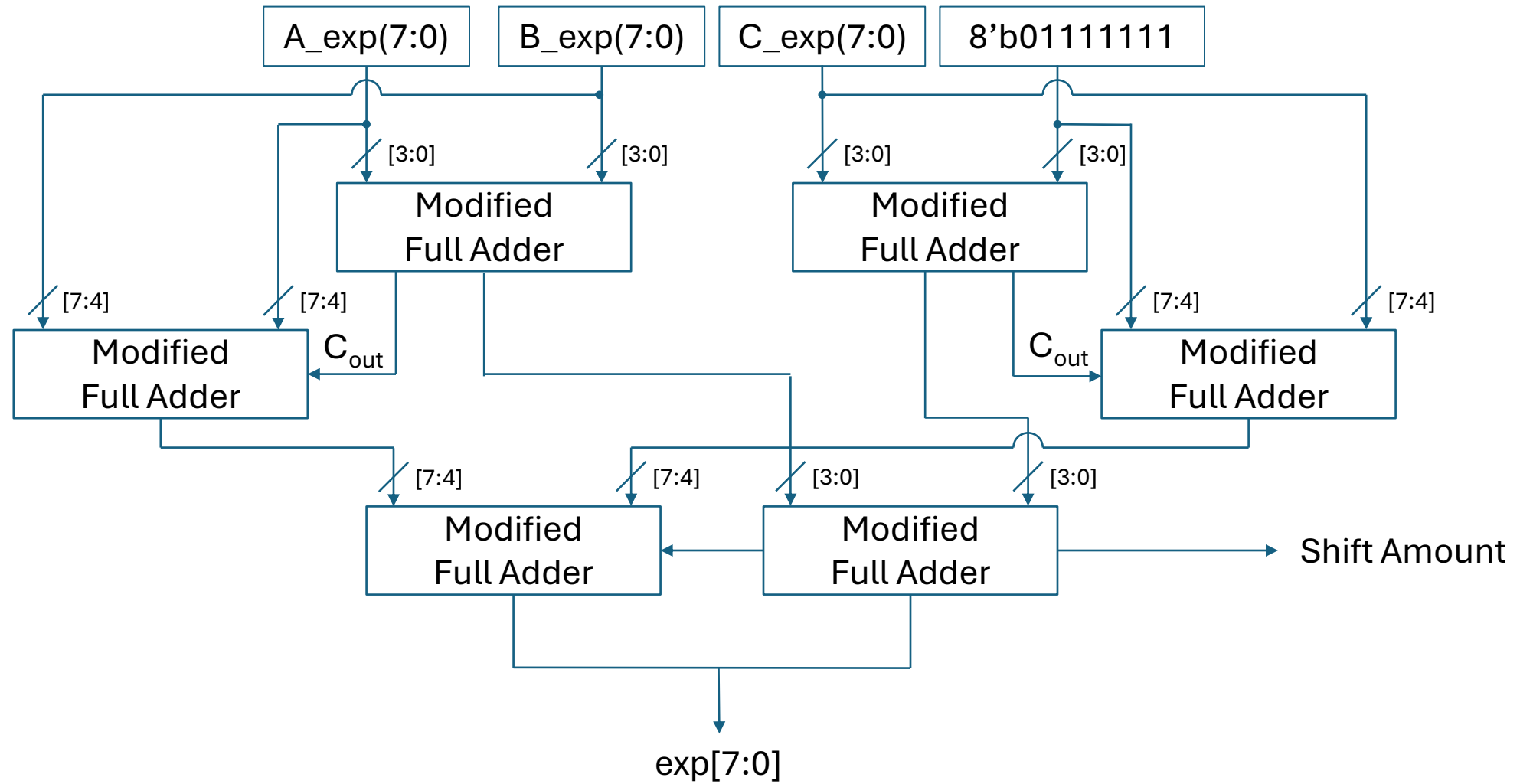
Unsigned 8-bit Multiply

																A ₇	A ₆	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
																x	B ₇	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀
																	P ₇₀	P ₆₀	P ₅₀	P ₄₀	P ₃₀	P ₂₀	P ₁₀	P ₀₀
															P ₇₁	P ₆₁	P ₅₁	P ₄₁	P ₃₁	P ₂₁	P ₁₁	P ₀₁		
														P ₇₂	P ₆₂	P ₅₂	P ₄₂	P ₃₂	P ₂₂	P ₁₂	P ₀₂			
													P ₇₃	P ₆₃	P ₅₃	P ₄₃	P ₃₃	P ₂₃	P ₁₃	P ₀₃				
												P ₇₄	P ₆₄	P ₅₄	P ₄₄	P ₃₄	P ₂₄	P ₁₄	P ₀₄					
											P ₇₅	P ₆₅	P ₅₅	P ₄₅	P ₃₅	P ₂₅	P ₁₅	P ₀₅						
										P ₇₆	P ₆₆	P ₅₆	P ₄₆	P ₃₆	P ₂₆	P ₁₆	P ₀₆							
									P ₇₇	P ₆₇	P ₅₇	P ₄₇	P ₃₇	P ₂₇	P ₁₇	P ₀₇								
S ₁₅	S ₁₄	S ₁₃	S ₁₂	S ₁₁	S ₁₀	S ₀₉	S ₀₈	S ₀₇	S ₀₆	S ₀₅	S ₀₄	S ₀₃	S ₀₂	S ₀₁	S ₀₀									

Source: “Design and implementation of an eight bit multiplier using Twin Precision technique and Baugh-Wooley algorithm” [9]

Donald McCauley TechAnalytics LLC January 5, 2026

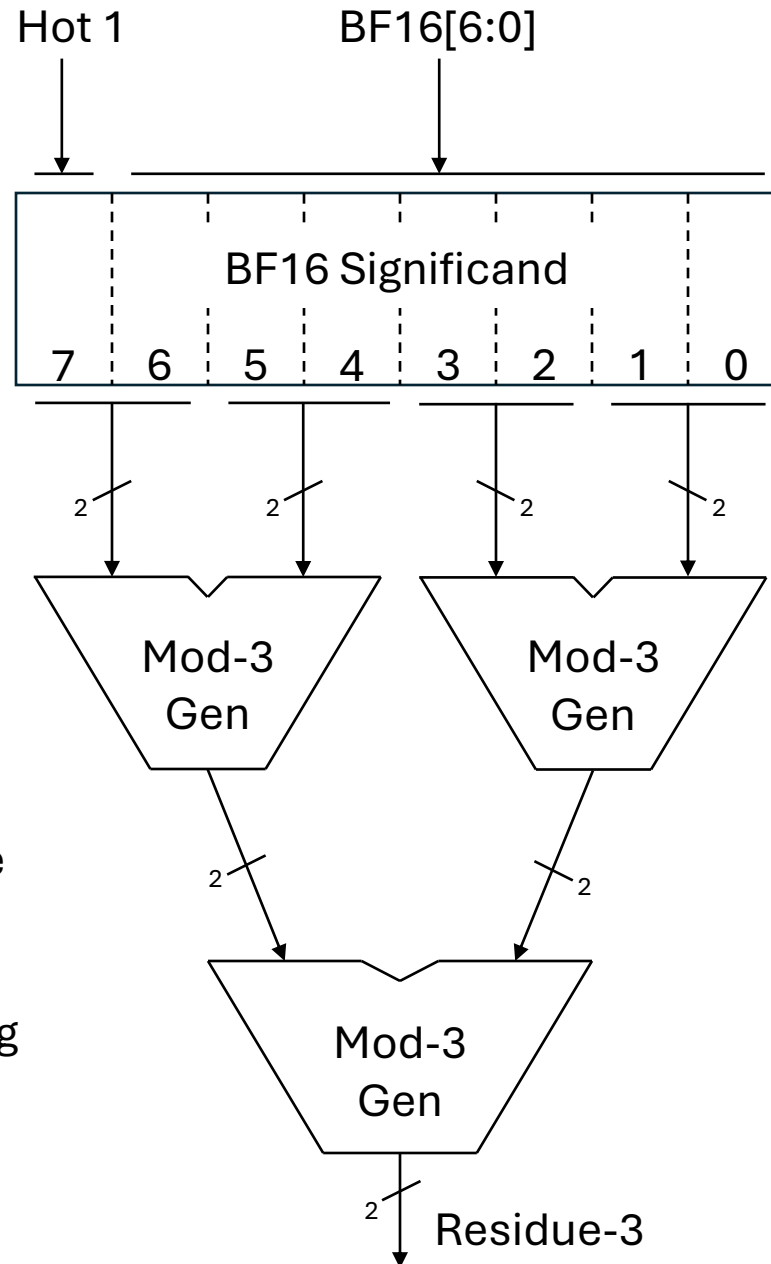
Exponent Difference / Adjust Logic



Source: 'A Bfloat16 Fused Multiplier-Adder' [7]

Residue-3 checking detects 100% of single bit errors and 66% of multiple bit errors.

The predicted residue of the product of two BF16 significands is simply the product of the corresponding residues.



BF16 Residue-3 Generation

Estimated cost in gates & delay:
~12 gates per Mod-3 generator;
~4 logic levels per Mod-3 generator.

A total of ~13 MOD-3 generators are required to generate residues for both 8-bit inputs as well as their 16-bit product. (Logic can be combined with input Residue-3 checking.)

Total cost of residue checking: TBD

Modulo-3 Generator

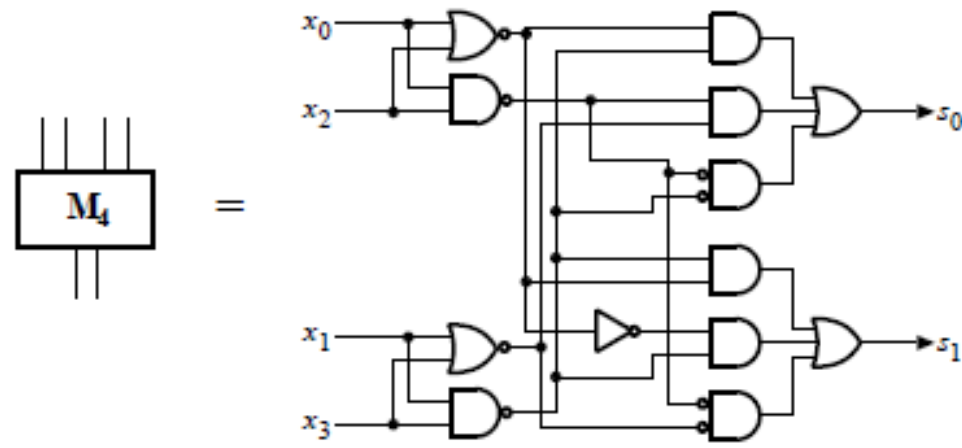


Figure 2: Cell M_4 of the 4-input generator mod 3.

Source: “VLSI IMPLEMENTATION AND COMPLEXITY COMPARISON OF RESIDUE GENERATORS MODULO 3” [8]

Backup

References:

- [1] D. McCauley, "The Case For High Availability GPUs", LinkedIn post, June 28, 2024
<https://www.linkedin.com/posts/activity-7212483952551297029-UJDv/>
- [2] S. Iacobovici, "End-to-end residue based protection of an execution pipeline", U.S. Patent 7,555,692, June 30, 2009
<https://patents.google.com/patent/US7555692B1/en>
- [3] R. W. Watson and C. W. Hastings, "Self-checked computation using residue arithmetic", Proceedings of the IEEE, vol. 54, no. 12, Dec. 1966
<https://ieeexplore.ieee.org/abstract/document/1447205>
- [4] Omid, R., Towhid, A., & Mohammadi, K., "A Survey on the Best Choice for Modulus of Residue Code", Indonesian Journal of Electrical Engineering and Informatics (IJEI), Dec 2019
<https://pdfs.semanticscholar.org/9455/4a4a613eb24a65426112e788f4f119f6b99e.pdf>
- [5] D. Lipetz and E. Schwarz, "Self Checking in Current Floating-Point Units," *2011 IEEE 20th Symposium on Computer Arithmetic*, Tuebingen, Germany, 2011
https://people.ee.duke.edu/~sorin/papers/itc06_multiplier.pdf
- [6] P. Patali and S. T. Kassim, "An Efficient Architecture for Signed Carry Save Multiplication," in *IEEE Letters of the Computer Society*, vol. 3, no. 1, 1 Jan.-June 2020
<https://ieeexplore.ieee.org/document/8979361>
- [7] S. Hutchins and E. Swartzlander, "A Bfloat16 Fused Multiplier-Adder," 2020 11th IEEE Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON), New York, NY, USA, 2020 ✓
<https://ieeexplore.ieee.org/document/9298120>
- [8] S. J. Piestrak, F. Pedron and O. Sentieys, "VLSI implementation and complexity comparison of residue generators modulo 3," *9th European Signal Processing Conference (EUSIPCO 1998)*, Rhodes, Greece, 1998 ✓
<https://ieeexplore.ieee.org/document/7089689>

References (Cont.):

- [9] R. K. Mallavarapu, T. S. Rao and S. Vasavi, "Design and Implementation of an Eight Bit Multiplier Using Twin Precision Technique and Baugh-Wooley Algorithm", *International Journal of Scientific and Research Publications (IJSRP)*, April 2013
<https://www.ijsrp.org/research-paper-0413/ijsrp-p1695.pdf>
- [10] Suman, Shruti & Singh, Ngangbam & Raja, Selvakumar & Saini, Harshita. (2021). "Design of 32-bit cell-based carry-save combinational multiplier with reduced area and propagation delay". *Journal of Physics: Conference Series*.
https://www.researchgate.net/publication/356715472_Design_of_32-bit_cell-based_carry-save_combinational_multiplier_with_reduced_area_and_propagation_delay
- [11] D. Eun Shim, P. Kumar, A. Ashok Kini, M. Mallikarjuna, M. Nahid Haque Shazon and A. Naeemi, "GT3: An Open-Source 3-nm GAAFET PDK and Platform for End-to-End Evaluation of Emerging Technologies," in *IEEE Transactions on Electron Devices*, vol. 72, no. 4, April 2025.
https://www.researchgate.net/publication/389415622_GT3_An_Open-Source_3-nm_GAAFET_PDK_and_Platform_for_End-to-End_Evaluation_of_Emerging_Technologies